Cybersecurity Home Lab Portfolio

Network Traffic Analysis — Wireshark

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Certifications	CompTIA A+ | Network+ | Security+
Platform	VMware Workstation 25H2
Tools	Wireshark 4.6.4, nmap 7.98, Kali Linux 2026.1
Capture NIC	VMware Network Adapter VMnet1 (Host-only)
Date	April 2026

Lab Environment

All exercises were performed in a fully isolated VMware Host-Only network (VMnet1). No VM has internet access, ensuring safe execution of offensive techniques against intentionally vulnerable targets.

Machine	Role / IP
Kali Linux 2026.1	192.168.128.130 — Attacker machine
Metasploitable 2	192.168.128.129 — Vulnerable target
Windows 11 x64	192.168.128.131 — Victim workstation
Host (Wireshark)	VMnet1 adapter — Packet capture

Screenshot 1 — VMware library showing all VMs configured on Host-only network

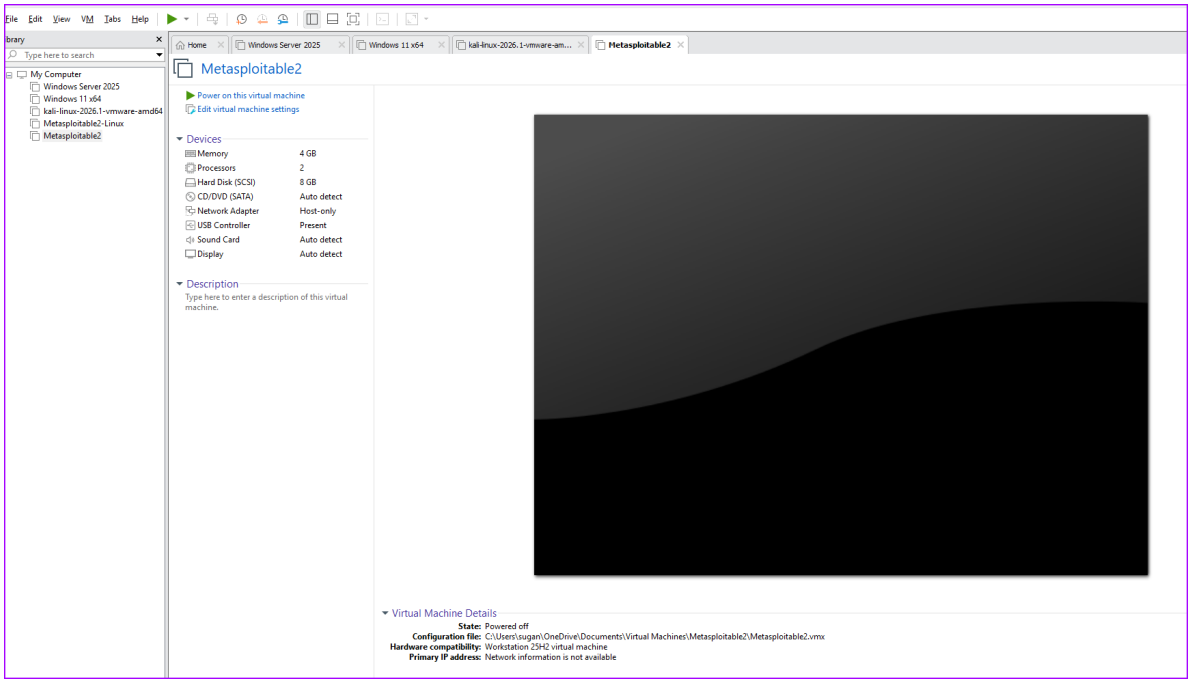


Fig 1. Metasploitable2 VM settings — Host-only network adapter confirmed

Screenshot 2 — Kali Linux IP configuration (192.168.128.130)

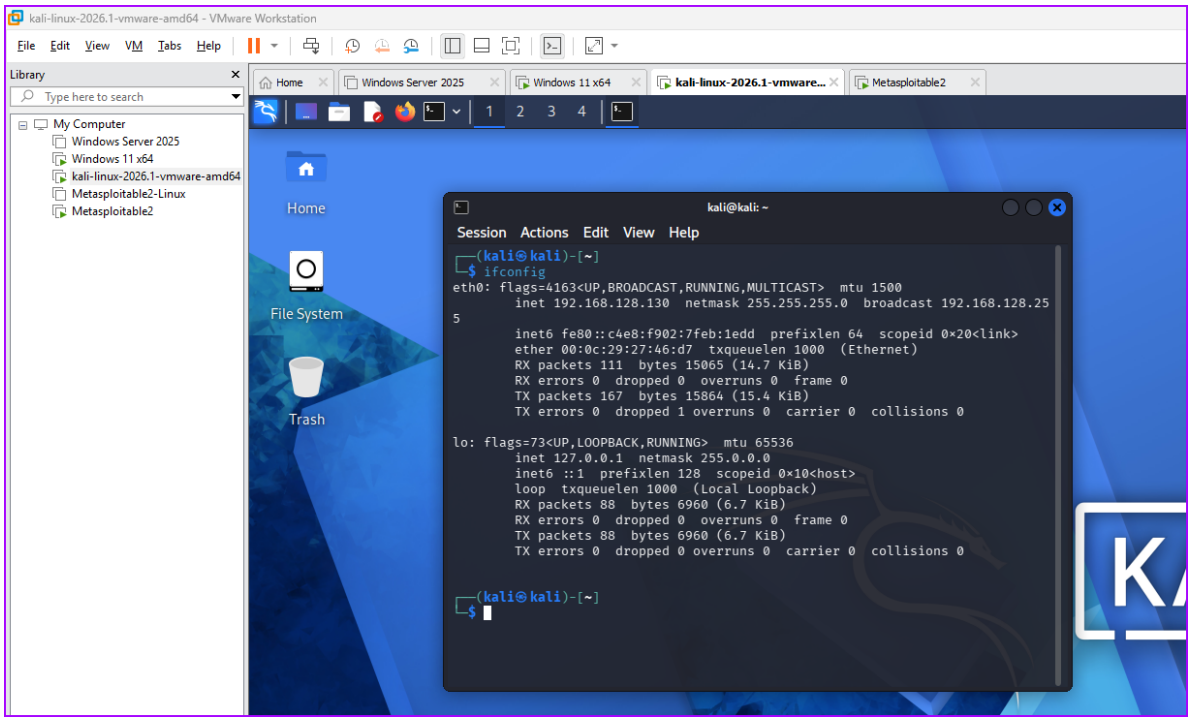


Fig 2. Kali Linux ifconfig output showing eth0 at 192.168.128.130

Screenshot 3 — Metasploitable2 IP configuration (192.168.128.129)

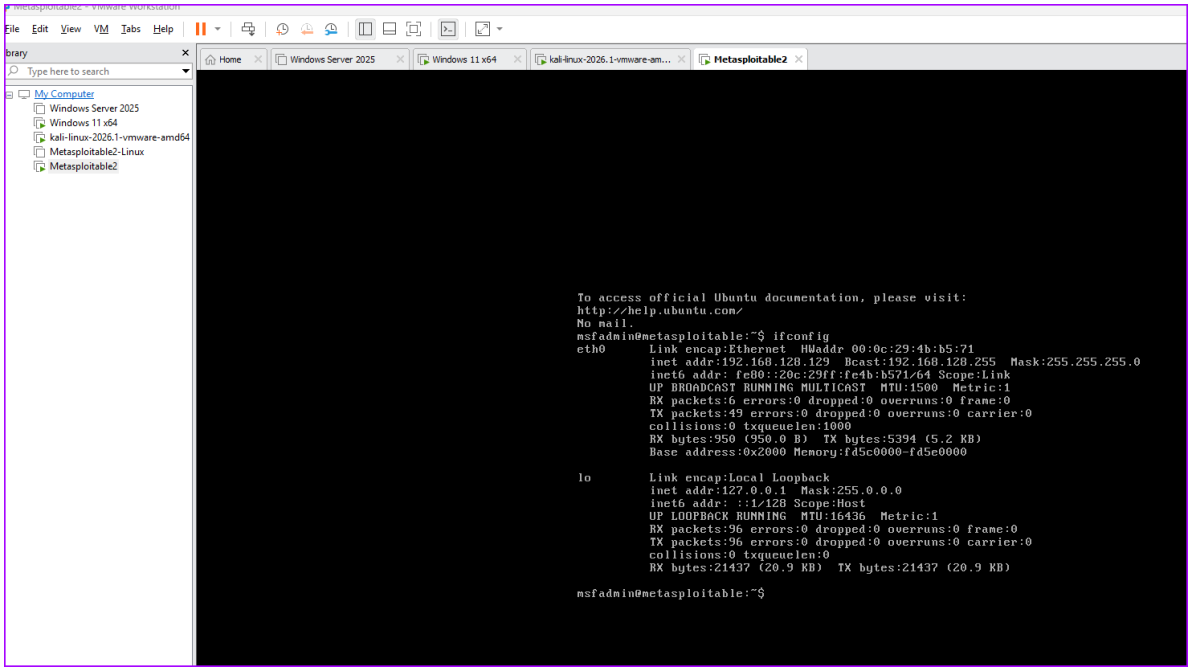


Fig 3. Metasploitable2 ifconfig output showing eth0 at 192.168.128.129

Objective

Detect and analyse a TCP SYN port scan performed by Kali Linux against Metasploitable2. Identify the network signature, enumerate discovered services, and document indicators of compromise.

Attack Simulation

Two nmap scans were executed from Kali Linux:

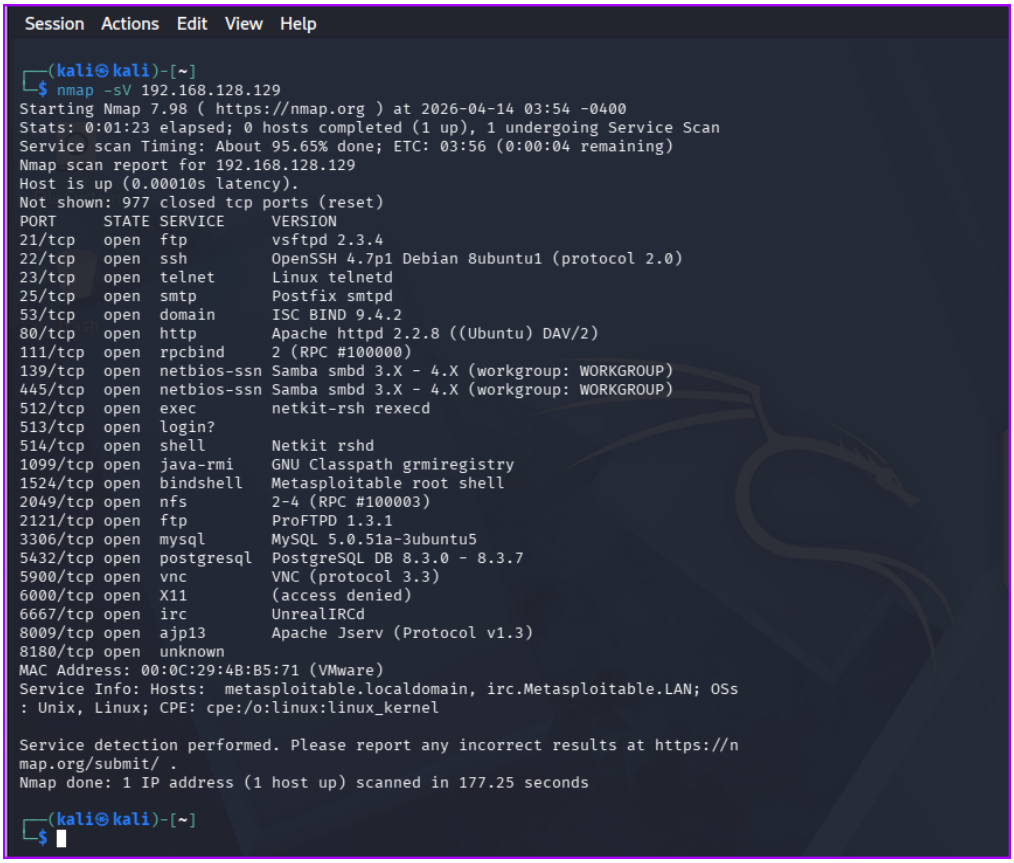
- **SYN scan** — half-open scan, does not complete the TCP handshake:

```
nmap -sS 192.168.128.129
```

- **Service version scan** — detects running service versions:

```
nmap -sV 192.168.128.129
```

Screenshot 4 — nmap -sV results showing open ports and service versions



```
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ nmap -sV 192.168.128.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-14 03:54 -0400
Stats: 0:01:23 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 95.65% done; ETC: 03:56 (0:00:04 remaining)
Nmap scan report for 192.168.128.129
Host is up (0.00010s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login?
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  unknown
MAC Address: 00:0C:29:4B:B5:71 (VMware)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs
: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 177.25 seconds

┌──(kali㉿kali)-[~]
└─$
```

Fig 4. nmap service scan reveals 20+ open ports including FTP (vsftpd 2.3.4), SSH, Telnet, HTTP, SMB, MySQL

Services Discovered on Metasploitable2

	Version	Risk
	vsftpd 2.3.4	Critical — C
	OpenSSH 4.7p1	Medium — c

	Linux telnetd	High — plain
	Postfix smtpd	Medium
	Apache 2.2.8	High — DVW
	Samba 3.x–4.x	Critical — kn
	5.0.51a	High — no r
	8.3.0–8.3.7	Medium
	Protocol 3.3	High — wea
	UnrealIRCd	Critical — ba

Detection in Wireshark

Wireshark was running on the host machine capturing all traffic on VMnet1 during the scan. The SYN scan produces a distinctive burst of TCP SYN packets from one source IP to hundreds of destination ports in milliseconds.

Display filter applied:

```
tcp.flags.syn==1 && tcp.flags.ack==0
```

Screenshot 5 — Wireshark live capture showing SYN scan traffic (unfiltered)

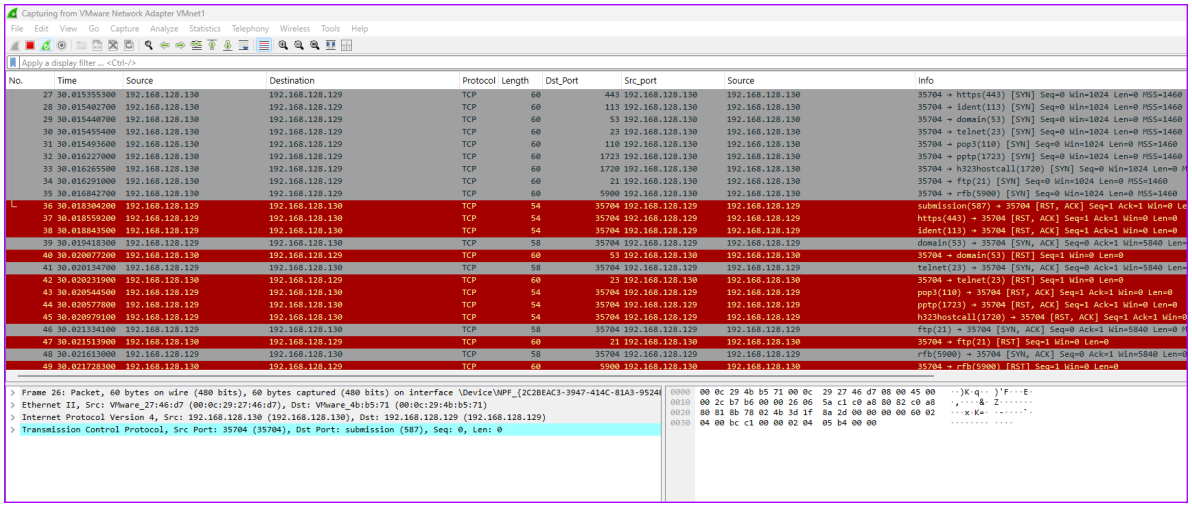


Fig 5. Live capture — Kali (192.168.128.130) sending SYN packets to Metasploitable (192.168.128.129) across multiple ports

Screenshot 6 — Filtered view: only SYN packets (tcp.flags.syn==1 && tcp.flags.ack==0)

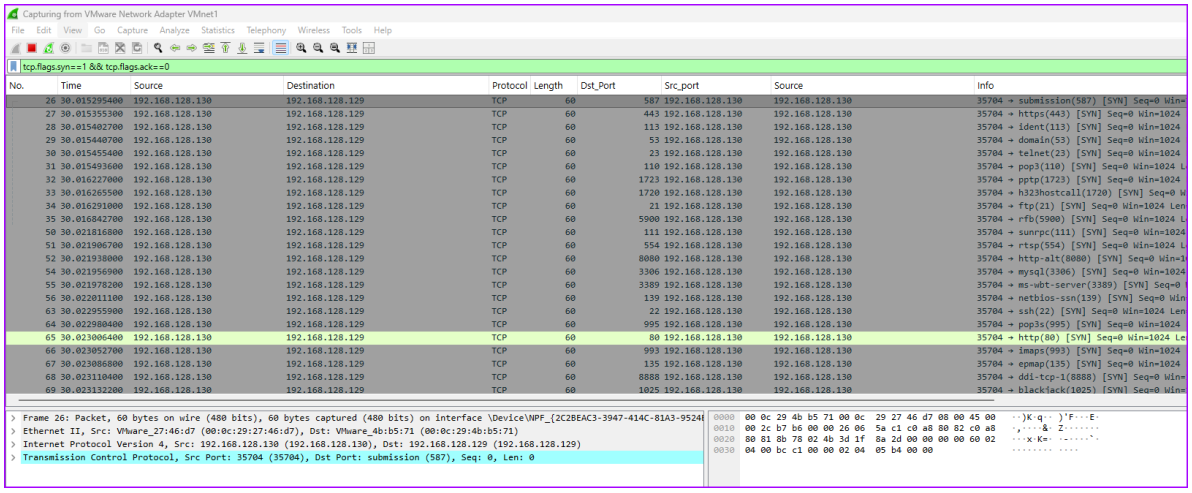


Fig 6. After applying SYN filter — all packets show [SYN] flag only, confirming half-open scan technique

Screenshot 7 — ICMP ping connectivity test between VMs

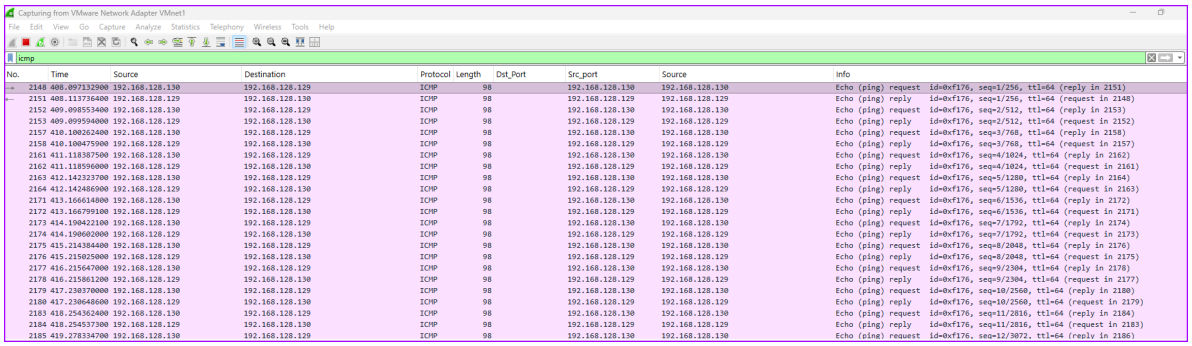


Fig 7. ICMP echo requests/replies confirming all three VMs are reachable on the lab network

Screenshot 8 — Statistics → Conversations TCP tab (1000+ port conversations)

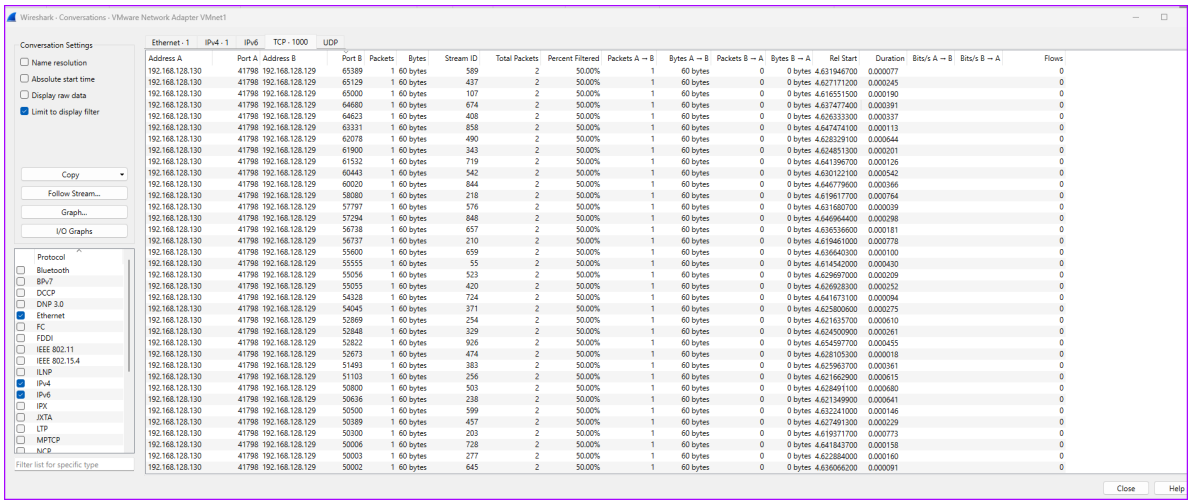


Fig 8. Conversations window showing 192.168.128.130 contacted 1000+ unique destination ports — classic port scan signature

Findings — Indicators of Compromise

Indicator	Value
Attacker IP	192.168.128.130 (Kali Linux)
Target IP	192.168.128.129 (Metasploitable2)

Scan type	TCP SYN half-open scan
Ports probed	1–65389 (1000+ unique ports)
Packet size	60 bytes (consistent — tool fingerprint)
Time between pkts	Milliseconds — automated scanning
Tool fingerprint	nmap 7.98 default timing
Duration	~177 seconds (service scan)

Recommendations

- Deploy Suricata or Snort IDS with SYN flood detection rules
- Rate-limit inbound SYN packets per source IP at the firewall (e.g. max 50 SYN/sec)
- Alert on any host contacting more than 50 unique ports within 10 seconds
- Disable all unused services on production hosts (Telnet, VNC, IRC, rsh)
- Patch vsftpd — version 2.3.4 contains a known backdoor (CVE-2011-2523)
- Enforce network segmentation so internal hosts cannot scan each other

Objective

Demonstrate that credentials transmitted over unencrypted protocols (FTP) are fully visible to a network observer. Capture the plaintext username and password using Wireshark and document the attack technique and remediation.

Attack Simulation

An FTP session was initiated from Kali Linux to Metasploitable2 using the default credentials. Wireshark captured all traffic on VMnet1 during the session.

```
ftp 192.168.128.129
```

Login credentials used:

Indicator	Value
Username	msfadmin
Password	msfadmin
Protocol	FTP (port 21) — plaintext, no encryption
Server	vsftpd 2.3.4 on Metasploitable2

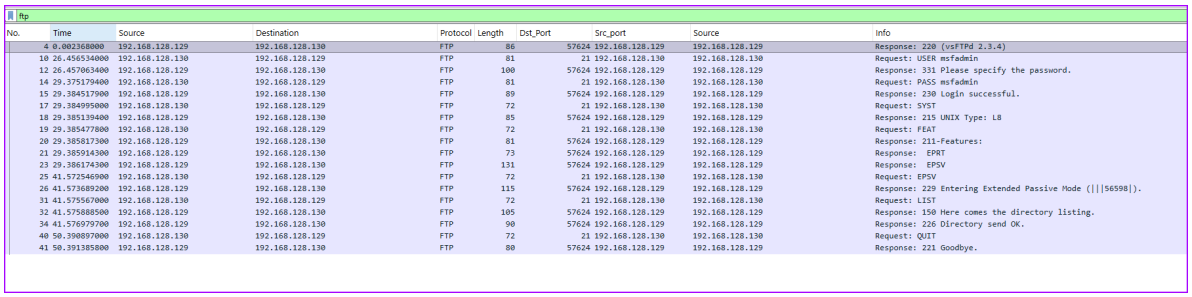
Detection in Wireshark

Display filter applied:

```
ftp
```

After filtering, right-clicking any FTP packet and selecting Follow → TCP Stream reveals the entire session including credentials in plaintext.

Screenshot 9 — FTP packet list with 'ftp' filter applied



ftp

No.	Time	Source	Destination	Protocol	Length	Dst_Port	Src_port	Source	Info
4	0.002368000	192.168.128.129	192.168.128.130	FTP	86	57624	192.168.128.129	192.168.128.129	Response: 220 (vsFTPd 2.3.4)
10	26.456534000	192.168.128.130	192.168.128.129	FTP	81	21	192.168.128.130	192.168.128.130	Request: USER msfadmin
12	26.457063400	192.168.128.129	192.168.128.130	FTP	100	57624	192.168.128.129	192.168.128.129	Response: 331 Please specify the password.
14	29.375179400	192.168.128.130	192.168.128.129	FTP	81	21	192.168.128.130	192.168.128.130	Request: PASS msfadmin
15	29.384517900	192.168.128.129	192.168.128.130	FTP	89	57624	192.168.128.129	192.168.128.129	Response: 230 Login successful.
17	29.384995000	192.168.128.130	192.168.128.129	FTP	72	21	192.168.128.130	192.168.128.130	Request: SYST
18	29.385139400	192.168.128.129	192.168.128.130	FTP	85	57624	192.168.128.129	192.168.128.129	Response: 215 UNIX Type: L8
19	29.385477800	192.168.128.130	192.168.128.129	FTP	72	21	192.168.128.130	192.168.128.130	Request: FEAT
20	29.385817300	192.168.128.129	192.168.128.130	FTP	81	57624	192.168.128.129	192.168.128.129	Response: 211-Features:
21	29.385914300	192.168.128.129	192.168.128.130	FTP	73	57624	192.168.128.129	192.168.128.129	Response: EPRT
23	29.386174300	192.168.128.129	192.168.128.130	FTP	131	57624	192.168.128.129	192.168.128.129	Response: EPSV
25	41.572546900	192.168.128.130	192.168.128.129	FTP	72	21	192.168.128.130	192.168.128.130	Request: EPSV
26	41.573689200	192.168.128.129	192.168.128.130	FTP	115	57624	192.168.128.129	192.168.128.129	Response: 229 Entering Extended Passive Mode (|||56598|).
31	41.575567000	192.168.128.130	192.168.128.129	FTP	72	21	192.168.128.130	192.168.128.130	Request: LIST
32	41.575888500	192.168.128.129	192.168.128.130	FTP	105	57624	192.168.128.129	192.168.128.129	Response: 150 Here comes the directory listing.
34	41.576979700	192.168.128.129	192.168.128.130	FTP	90	57624	192.168.128.129	192.168.128.129	Response: 226 Directory send OK.
40	50.390897000	192.168.128.130	192.168.128.129	FTP	72	21	192.168.128.130	192.168.128.130	Request: QUIT
41	50.391385800	192.168.128.129	192.168.128.130	FTP	80	57624	192.168.128.129	192.168.128.129	Response: 221 Goodbye.

Fig 9. FTP filter showing USER msfadmin and PASS msfadmin visible in the Info column

Screenshot 10 — Full FTP session with TCP/FTP packets (unfiltered)

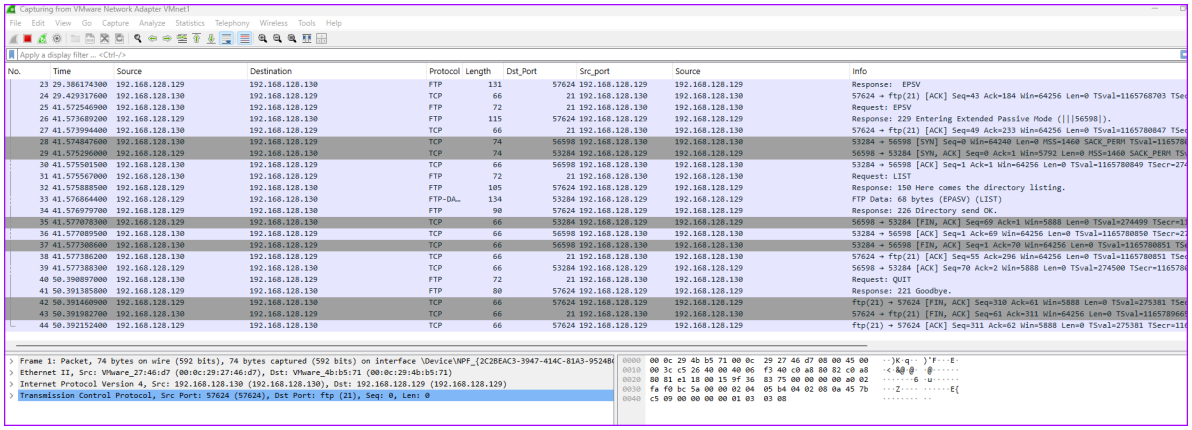


Fig 10. Complete FTP session capture showing connection establishment, data transfer, and session teardown

Screenshot 11 — Follow TCP Stream showing plaintext credentials

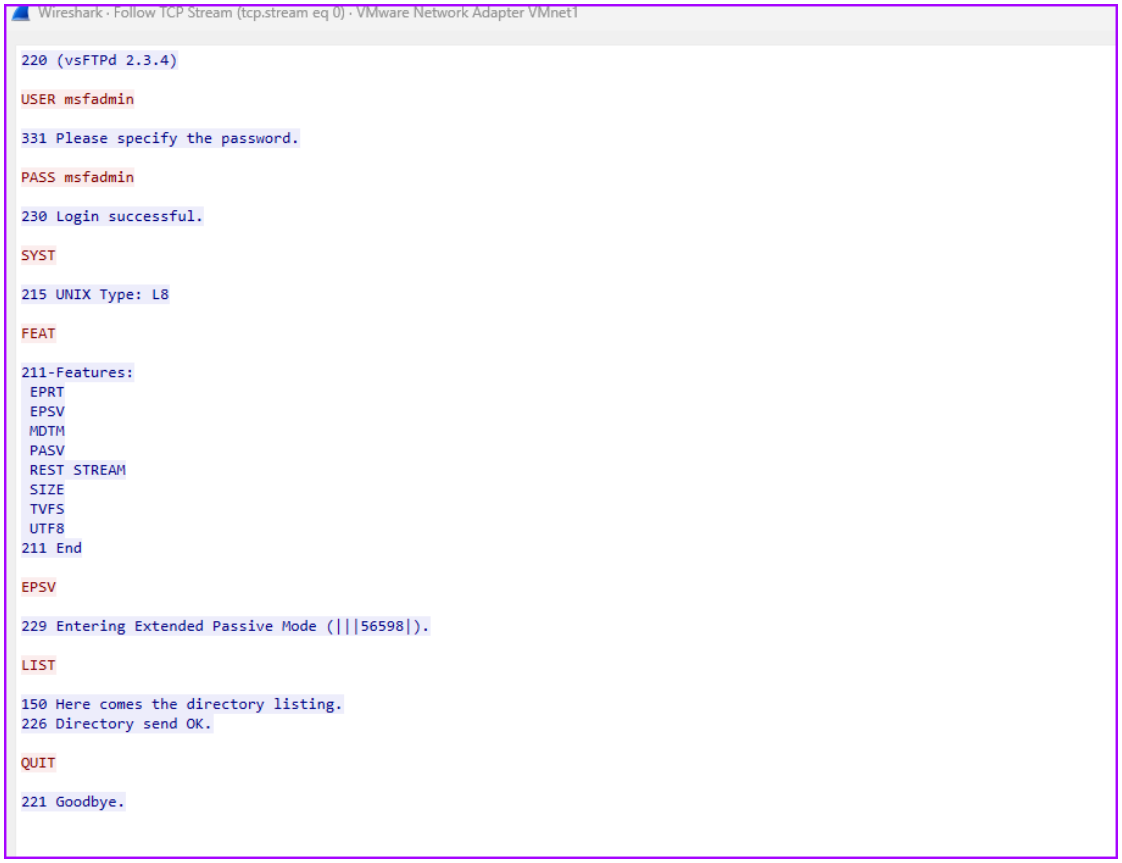


Fig 11. TCP Stream — USER msfadmin and PASS msfadmin fully visible in red (client-sent data)

TCP Stream Analysis

Client (red)	Server (blue)
`USER msfadmin`	`331 Please specify the password.`
`PASS msfadmin`	`230 Login successful.`
`SYST`	`215 UNIX Type: L8`
`FEAT`	`211-Features: EPRT EPSV MDTM...`
`EPSV`	`229 Entering Extended Passive Mode`

`LIST`	`150 Here comes the directory listing.`
`QUIT`	`221 Goodbye.`

Findings — Indicators of Compromise

Indicator	Value
Attacker IP	192.168.128.130 (Kali Linux)
Target IP	192.168.128.129 (Metasploitable2)
Protocol	FTP — port 21
Captured username	msfadmin
Captured password	msfadmin
Server software	vsftpd 2.3.4
CVE	CVE-2011-2523 (vsftpd 2.3.4 backdoor on port 6200)
Encryption	None — fully plaintext transmission

Why This Is Dangerous

FTP transmits all data including credentials in plaintext. Any attacker positioned between client and server — via ARP spoofing, a compromised switch, or a rogue access point — can capture valid credentials silently without the user knowing. These credentials can then be reused across other services (credential stuffing attack).

Recommendations

- Disable FTP entirely — replace with SFTP (SSH File Transfer Protocol) or FTPS
- Enforce TLS/SSL on all file transfer protocols at the network policy level
- Never reuse credentials across services — use a password manager
- Implement 802.1X port authentication to prevent unauthorised network taps
- Patch vsftpd immediately — version 2.3.4 contains a backdoor accessible on port 6200
- Deploy network monitoring to alert on any FTP traffic in production environments

Skills Demonstrated	
Skill	**Evidence**
Network traffic capture	Wireshark on VMnet1 — host-based capture of VM traffic
Display filter proficiency	tcp.flags.syn==1, ftp, icmp filters applied correctly
Attack simulation	nmap SYN/service scans, FTP credential sniffing
IOC identification	Attacker IPs, credentials, CVEs, tool fingerprints
Threat analysis	Explained mechanism of each attack technique
Remediation advice	Specific, actionable, technically accurate
Lab environment setup	VMware, 3-VM isolated host-only network
Documentation	Professional report with evidence screenshots

Next Projects (Planned)

- C2 beaconing simulation — Metasploit Meterpreter reverse shell, IO Graph analysis
- DNS tunneling detection — high-volume DNS queries, oversized query detection
- Lateral movement via SMB — pass-the-hash, SMB enumeration between VMs
- Suricata IDS — correlate alerts with Wireshark captures for the same events

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This portfolio was built in an isolated VMware lab environment. All attacks were performed against intentionally vulnerable machines. No real systems were involved.